nature portfolio

Peer Review File

Patterns of partisan toxicity and engagement reveal the common structure of online political communication across countries



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Reviewers' Comments:

Reviewer #1:
Remarks to the Author:
Thank you for the opportunity to review "Affective and interactional polarization align across countries" for Nature Communications. The authors have put together an extensive study that effectively links affective polarization with interactional polarization across nine diverse countries. This work shines a spotlight on the impact of polarization within social media discourse.
Firstly, I'd like to commend the authors' ambitious endeavor, which thoughtfully extends beyond the common US-centric scope. It's important that we encourage and recognize such global, meticulously conducted research. Consequently, I firmly believe that this manuscript's findings are valuable and merit publication in Nature Communications.
Approaching the review with a positive mindset, I would like to offer some constructive feedback to further enhance the manuscript before recommending it for acceptance.
The main area where the paper could be improved is in creating a more cohesive narrative. The paper contains numerous interesting analyses; however, they appear to be slightly disconnected. A stronger thread is needed to weave them together systematically. Although the descriptive nature of the study is appreciated, a more detailed explanation and theoretical foundation could help stitch together the progression from toxicity to engagement and misinformation. Clarity in this regard could be significantly improved by drawing on existing theories and models of polarization in social media. Here are my specific suggestions:
1. The intersection of polarization, incivility, and engagement might be better explained with reference to existing theories and models. I refer the authors to the works of William Brady, MJ Crocket and others for example.

Brady, W.J., Crockett, M.J., and Van Bavel, J.J. (2020). The MAD Model of Moral Contagion: The Role of Motivation, Attention, and Design in the Spread of Moralized Content Online. Perspect. Psychol. Sci. 15, 978–1010.

Brady, W.J., Jackson, J.C., Lindström, B., and Crockett, M.J. (2023). Algorithm-mediated social learning in online social networks. Trends Cogn. Sci. 10.1016/j.tics.2023.06.008.

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2. I recommend incorporating discussions and citations from other studies that investigate the connections between interactional polarization, in-group/out-group engagement, and incivility, like: Brady, W.J., Wills, J.A., Jost, J.T., Tucker, J.A., and Van Bavel, J.J. (2017). Emotion shapes the diffusion of moralized content in social networks. Proceedings of the National Academy of Sciences, 114, 7313–7318.
3. The analysis regarding news quality is certainly intriguing and amplifies findings from other studies (e.g., Lasser et al., 2022; PNAS Nexus). However, it might be beneficial to relegate this to supplementary material to avoid detracting from a more focused narrative centered on

interactional polarization, toxicity, and engagement.

4. Upon evaluating the current analyses, a logical progression would be to examine toxic engagement among both politically engaged and apolitical users. Adding this extra layer of analysis could greatly contribute to the cohesiveness of the paper. A hypothesis that might be explored is whether toxicity correlates with engagement differently amongst these two groups, a phenomenon which, when elucidated, could enrich the manuscript's insights.

5. The discussion section touches on an important concept—the notion of echo chambers—which might be worthy of more in-depth exploration. It would be beneficial if this could be expanded upon, perhaps by differentiating between "echo chambers," influenced by homophily, and "filter bubbles," induced by feed algorithms.

In closing, I am excited about the potential impact of this research and look forward to seeing how it can be refined to tell a story that's not only informative but also compelling and coherent. I believe with these suggested improvements, the manuscript will be well-suited for publication.

Sincerely,
Almog Simchon

Reviewer #2:
Remarks to the Author:
This uses a large Twitter dataset to investigate the extent to which affective and interactional polarization align in several countries. The work is very relevant, as this is still an open question, and the methodology is mostly sound. There are several interesting and new results on the toxicity and engagement within and across ideologies, however, there are also several limitations to their results, such as the fact they studied only 24h of Twitter activity and only looked at activity related to elected politicians. As the authors recognize, this is a first step to understanding this issue better. However, it is not definitive evidence showing that affective and interactional polarization are always aligned. For this reason, in its present form, the paper may fall just a bit short in terms of the generalization of its results.

I have two main issues:
1) The authors decided to focus only on tweets related to elected politicians, but I expect that a lot of political conversations occur without mentioning or retweeting politicians. For example, it would have made sense to include tweets from media outlets as well. Could the authors repeat their analysis by including the most prominent media outlets in each country or at least provide a clear reason for not explicitly including them?

2) About the ideology scores extraction. This is where I have some questions about the methodology.
First, the authors claim that they find two opposing groups in each country, and much of the

analysis relies on this. But, they actually force the ideology to be one-dimensional. The ideology is extracted from an SVD decomposition of a rescaled bipartite incidence matrix, and only the first dimension is used. Here, the authors could look at the other dimensions to see if they have significant information (the corresponding singular value should give an idea of the importance of the dimension). In particular, looking at Fig. 2. it seems that France, Germany, Poland and others may be better captured with more dimensions.
Then there is the question of the rescaling using the two largest peak. There could be other ways to scale the scores (e.g., by removing the mean, for example), which would change where the 0 value is and therefore impact which users are considered left or right. This probably would not affect the results too much, but some robustness checks would be good.

Smaller remarks:
- line 333: " Figure 3(a) shows boxplots of the bootstrapped difference in the median toxicity quantile of out-group interactions less the median toxicity quantile of in-group interactions for each country". This sentence is confusing. The median is a quantile, but I don't see what the median toxicity quantile is... Do the authors mean they looked at the ranks of posts sorted by toxicity? Maybe a mathematical definition would clarify things.
- for the engagement study, why did they use the mean log like-count and not the median, as for the toxicity?
- In the conclusion, "our results also reveal how partisan non-conformists are essentially treated as members of the political opposition." I wonder if this is not because they only used a 1-D ideology. Including more dimensions could capture the ideology of these non-conformists.

Reviewer #3:
Remarks to the Author:
The authors analyze a quite unique dataset comprising all public tweets and re-tweets in one day across countries. The authors start by applying an “unsupervised learning” technique to derive an ideological score for each user and cluster them based on interaction patterns with active politicians on Twitter. This ideological score is used to distribute users between two groups — political left and political right. Then, the authors move on to study the level of segregation revealed according to this group definition; the level of toxicity and engagement in in/out-group posts; the level of reliability in sources referenced by the two groups; and other characteristics of Twitter interactions, independent on group identity (e.g., political vs apolitical interactions and reciprocity in interactions).

The authors conclude that Twitter reveals high levels of interactional polarization: users belonging to different ideological group preferentially retweet politicians from different political parties. The authors conclude that out-group interactions are more toxic and less engaging (that is, receive less likes). Finally, the ideological group associated with the political right references less reliable media sources.

Overall, I found the analysis presented in this manuscript sound. I believe the conclusions are

original and interesting to a multidisciplinary community interested in dynamics of political discourse on social media platforms. Although the paper reports many findings, the results section follows a very clear structure, and it is easy to follow all steps of the analysis. My general assessment is positive. Nonetheless, I have some suggestions for improvement:

- The definition of "political" and "apolitical" content followed here ignores posts that might be politically loaded without mentioning politicians' accounts. The authors compute users' ideological score given their retweets of politician's accounts; however, users can express their views or endorse/criticize policies without explicitly referring to a politician. Alternative methods such as topic modelling could provide ways of classifying the ideological leaning of posts without resorting to mentions to the accounts of elected politicians. The authors should discuss eventual limitations of their definition of political content.

- The decision to ignore users that retweet more than 10 politicians (why 10?) and the decision to add use 2 retweets (why 2?) as the minimum threshold to add a link in the co-retweet network still seems rather arbitrary. The authors could provide extra data on the robustness of their conclusions if those two thresholds change.

- The dataset used is distinctive as it comprises all Twitter posts in a 24-hour period. At the same time, this 1-day window might pose limitations. September 21 2022 might represent a different day in terms of the electoral cycles and specific political events in different countries. As the authors discuss, the results for Germany are very much dependent on the government's position on the Russia-Ukraine war (would the same conclusions hold if the the day was September 21 2021 or September 21 2023?). It remains unclear how the results presented generalize to longer time-windows; the authors could perhaps discuss this.

Minor points:

- Line 657 and Figure 1 caption: It is mentioned that the network visualization is produced using a force directed drawing algorithm (ForceAtlas2). This might sound rather cryptic for readers not familiar with this algorithm. My suggestion is to provide an intuition for what this means in terms of attraction and repulsion between nodes.

- Line 693: "retweeted users" or "retweeted influencers"?

- Line 687: please clarify the dimensions of vectors r and c and the exact product implied by rc — this was confusing as r and c do not have the same dimensions.

- The ideological scores considered are continuous, yet they are collapsed onto a binary classification (left-right). The authors could comment potential limitations for this approach, especially in countries where many users have a score close to 0 (see France).

Reviewer #4:
Remarks to the Author:
I have read the paper with great interest. Across nine studies, the authors investigate whether political networks on twitter are polarized. They found evidence for bipartite networks in all countries, and for outgroup interactions being characterized by higher toxicity and less engagement.

I only have a few minor comments, mostly related to expanding the theoretical contribution of the findings:

1) In the political ideology literature, there is a debate on whether only rightist show discrimination (see e.g. Balliet, D., Tybur, J. M., Wu, J., Antonellis, C., & Van Lange, P. A. (2018). Political ideology, trust, and cooperation: In-group favoritism among Republicans and Democrats during a US national election. Journal of Conflict Resolution, 62(4), 797-818). The authors briefly mention these results but I suggest to provide more details on this potential asymmetry. Overall, do rightist vs leftist show for outgroup toxicity, or it is the other way around? Or they simply show the same levels of toxicity? Is there variation across countries?

2) The authors say that outgroup interactions are characterized by more toxicity and less engagement. Is the result on toxicity with ingroup (vs outgroup) in terms of proportion (i.e., i have less interactions with outgroup and this lower number of interactions is mostly toxic) or absolute values? In other words, one possibility is that ingroup interactions can also be toxic but at the same time they are also characterized by many non-toxic interactions. Can the authors speak to this?

3) The authors mention the UK as an outlier (as they show higher fraction of toxicity towards ingroup). Given the amount of data, i would actually suggest to give importance to this result. In a way, it show the strong limitation of only focusing on one country (the US). These findings show that outgroup toxicity is not a universal but at times, people can also be nasty with ingroup members.

4) Is there variation in outgroup toxicity and is this variation meaningful? Previous cross-cultural studies have tried to understand whether ingroup bias varies across societies (see Dorrough & Gloeckner, 2016, PNAS; or Romano et al., 2021, Nat Comms). I think it would enrich the paper to have some insights on whether outgroup hate meaningfully covary with prominent cross-societal factor (e.g., gdp, or rule of law).

Dear Reviewers,

Thank you very much for taking the time to read and review our manuscript. Your comments have been invaluable for improving the paper as a whole.

Below, you will find a detailed response to each of your comments including the line numbers where changes can be found in the revised manuscript. Some of the key changes include:

- The introduction and discussion have been significantly restructured and improved to better contextualize our work in the existing literature on affective polarization online, and cross-country differences between in- and out-group behavior.
- A new analysis of apolitical mentions has been added, showing that in all nine countries political mentions are more toxic than apolitical mentions. We have discussed in detail our definition of political / apolitical to maximize clarity for the reader.
- We have added a range of robustness checks including (1) an investigation of the second dimension of the latent ideology, (2) a check of the toxicity results using an alternate rescaling of the ideology scores, (3) greater detail on the structural position of political outliers, and (4) a check that our analysis of out-group toxicity exceeding in-group toxicity is robust using raw toxicity scores rather than quantiles.
- To reflect the breadth of results in the paper, and to ensure accessibility to a broad readership, we have updated the title of the paper to “Common patterns of political polarization on social media: Out-group animosity, network interactions, and user engagement”.

We hope you will agree that the paper is now much improved, and look forward to hearing from you.

Response to Reviewer 1

Reviewer Comment 1.1

Thank you for the opportunity to review "Affective and interactional polarization align across countries" for Nature Communications. The authors have put together an extensive study that effectively links affective polarization with interactional polarization across nine diverse countries. This work shines a spotlight on the impact of polarization within social media discourse.

Firstly, I'd like to commend the authors' ambitious endeavor, which thoughtfully extends beyond the common US-centric scope. It's important that we encourage and recognize such global, meticulously conducted research. Consequently, I firmly believe that this manuscript's findings are valuable and merit publication in Nature Communications.

Author Response 1.1

We are very grateful to the reviewer for their encouraging comments and constructive feedback. Thank you very much for your time and effort.

Reviewer Comment 1.2

Approaching the review with a positive mindset, I would like to offer some constructive feedback to further enhance the manuscript before recommending it for acceptance.
The main area where the paper could be improved is in creating a more cohesive narrative. The paper contains numerous interesting analyses; however, they appear to be slightly disconnected. A stronger thread is needed to weave them together systematically. Although the descriptive nature of the study is appreciated, a more detailed explanation and theoretical foundation could help stitch together the progression from toxicity to engagement and misinformation. Clarity in this regard could be significantly improved by drawing on existing theories and models of polarization in social media. Here are my specific suggestions:

1. The intersection of polarization, incivility, and engagement might be better explained with reference to existing theories and models. I refer the authors to the works of William Brady, MJ Crocket and others for example.

Brady, W.J., Crockett, M.J., and Van Bavel, J.J. (2020). The MAD Model of Moral Contagion: The Role of Motivation, Attention, and Design in the Spread of Moralized Content Online. Perspect. Psychol. Sci. 15, 978–1010.

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Author Response 1.2

We thank the reviewer for their positive outlook on our paper and for their suggestions on framing and overall narrative - the suggested references are particularly useful in this regard. In the revised manuscript we have made a concerted effort to improve the narrative structure of the paper by restructuring the introduction, and have placed the discussion of online toxicity in the context of the existing literature on moral-emotional language on social media and the role of this language in driving online engagement. These changes have been incorporated in the introduction, **see lines 89 - 108**, and in the discussion, **see lines 666 - 673** and **lines 680 - 682**. The references suggested by the reviewer have been included in the revised manuscript as **Refs. [44 - 46]**.

Reviewer Comment 1.3

2. I recommend incorporating discussions and citations from other studies that investigate the connections between interactional polarization, in-group/out-group engagement, and incivility, like:

> Brady, W.J., Wills, J.A., Jost, J.T., Tucker, J.A., and Van Bavel, J.J. (2017). Emotion shapes the diffusion of moralized content in social networks. Proceedings of the National Academy of Sciences, 114, 7313–7318.

Author Response 1.3

We thank the reviewer for pointing us towards this highly interesting study which we agree is of great relevance to our study. In the revised paper, we have contextualized our results using this paper and have discussed how our study, which focuses on toxic language, relates to the moral-emotional words studied by Brady et al. 2017, see **lines 89 - 108** and **Ref. [43]**. We have also cited other papers studying online incivility, and have discussed how engagement with toxic content may be suppressed, relative to non-toxic content, see **lines 103 - 108** and **lines 666 - 676** and **Refs. [47, 48]**.

Reviewer Comment 1.4

3. The analysis regarding news quality is certainly intriguing and amplifies findings from other studies (e.g., Lasser et al., 2022; PNAS Nexus). However, it might be beneficial to relegate this to supplementary material to avoid detracting from a more focused narrative centered on interactional polarization, toxicity, and engagement.

Author Response 1.4

We thank the reviewer for this useful suggestion. In the revised manuscript we have cited the Lasser et al. 2022 study, see **Ref. [73]**, and have moved our results to the supplement. These results are briefly referenced in the main text in **lines 623 - 634**. This change has allowed us to

focus in greater detail on the narrative surrounding polarization, toxicity, and engagement, as suggested by the reviewer.

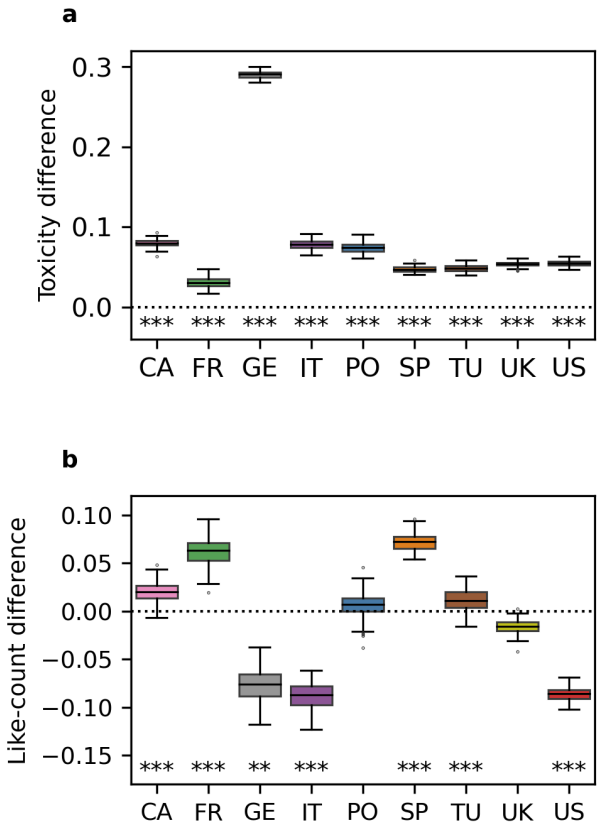
Reviewer Comment 1.5

4. Upon evaluating the current analyses, a logical progression would be to examine toxic engagement among both politically engaged and apolitical users. Adding this extra layer of analysis could greatly contribute to the cohesiveness of the paper. A hypothesis that might be explored is whether toxicity correlates with engagement differently amongst these two groups, a phenomenon which, when elucidated, could enrich the manuscript's insights.

Author Response 1.5

We thank the reviewer for this interesting suggestion which we are happy to incorporate into the new manuscript. To address this question, we have extracted all tweets with a unique mention to an apolitical account, i.e., those shown in Figure 5a who were mentioned by users with an assigned political ideology, but who do not have an ideological score themselves. This new analysis covers approximately 7 million tweets. The toxicity of these tweets is then measured using the methods described in the manuscript. Note that we include mentions authored both by users with an assigned political ideology, and authored by accounts without an ideological score – our focus is on who is mentioned, not on who authors the tweet.

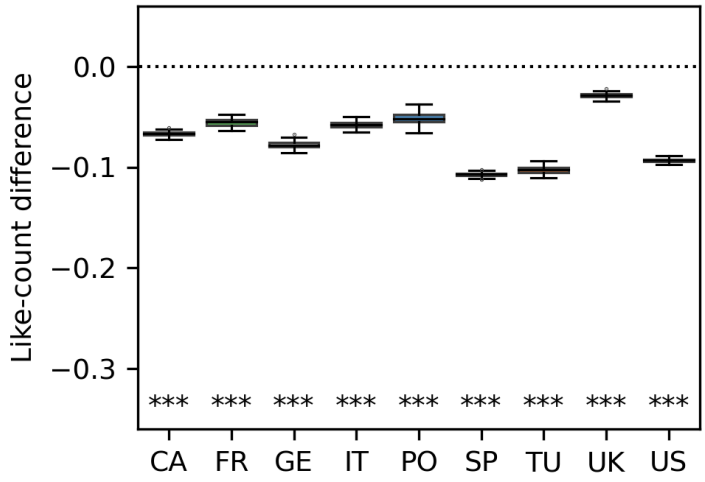
Having extracted these apolitical mentions, we first measure whether the toxicity of these tweets, and the engagement received, differs from the toxicity and engagement received by political mentions (i.e., those included in Figure 3). This is shown here:



Political mentions (i.e., mentioning an account with an assigned ideological score, either in-group or out-group) are more toxic than apolitical mentions (i.e., mentioning an account without an ideological score). There are no consistent cross-country trends as to whether political mentions, or apolitical mentions, receive more engagement. *(a) Boxplots for the bootstrapped difference between the median interaction toxicity between accounts with assigned ideological scores, less the median interaction toxicity for tweets mentioning an account without a known ideological score for each of the nine countries. (b) Boxplots for the bootstrapped difference between the mean log likes received by a post mentioning an account with an assigned ideological score, less the mean log likes received by posts mentioning an account without an assigned ideological score. Boxplots show the bootstrapped median, interquartile range (IQR), whiskers for 1.5 times the IQR from the hinge, and points for outliers (see Methods). Stars indicate statistical significance using the non-parametric Mann-Whitney U test on full distribution (see Methods): p<0.05: *, p< 0.01: **, p<0.001: ***.*

The above figure shows that political mentions are more toxic than mentions involving accounts without a known political ideology in all nine countries. There are no consistent trends regarding engagement. In Canada, France, Spain, and Turkey, political mentions receive greater engagement than apolitical mentions. In Germany, Italy, and the USA, apolitical mentions receive greater engagement than political mentions. No statistically significant difference is detected in either Poland or the UK.

We can also look at engagement with toxic content in general. Here we focus exclusively on apolitical mentions, and compare engagement with the most toxic mentions to engagement with all other apolitical mentions. This is equivalent to the original Figure 4, but now using apolitical mentions:



High toxicity interactions involving a mention of an apolitical account receive lower engagement than low toxicity interactions involving a mention of an apolitical account. *Boxplots show the bootstrapped mean log like-count received by high toxicity posts, minus the mean log like-count received by low toxicity posts. High toxicity interactions are defined as the top 10% most toxic posts in each country. Boxplot show the bootstrapped median, interquartile range (IQR), whiskers for 1.5 times the IQR from the hinge, and points for outliers (see Methods). Stars indicate statistical significance using the non-parametric Mann-Whitney U test on full distribution (see Methods): p<0.05: *, p< 0.01: **, p<0.001: ***.*

This figure shows that apolitical mentions follow the same engagement trend as political mentions. In both cases, high toxicity posts receive lower engagement than low toxicity posts.

These new results have been added to the main manuscript as the new **Figure 6** and to the supplementary material in **Figure S10**. The results are discussed in **lines 574 - 604** of the main manuscript.

Importantly, note that these results compare political mentions to apolitical mentions. However, it is critical that we clarify what we mean by the term apolitical in the current context. While the interactions focused on in the main paper involving elected politicians are self-evidently political, not retweeting an elected politician does not mean a user does not have an ideological leaning. Consequently, our cohort of “apolitical” accounts includes a large number of mentions which do not discuss political topics (they may, for instance, relate to entertainment, see **Ref. [61]**). However, this cohort also includes some accounts who are political but who did not engage with an elected politician during our observation window. Given the size of our dataset and the nuances as to what is and is not considered political, we cannot comprehensively separate these interactions other than by using our current method of focusing on interactions with elected politicians. But, considering that this cohort includes a mix of accounts from various backgrounds and covering diverse topics, it is reasonable to refer to this cohort as apolitical in a relative sense

when compared to the explicitly political interactions involving users who retweet elected politicians. To avoid any confusion on the part of the reader, we have explained this language in detail in the revised manuscript, see **lines 538 - 545**.

Reviewer Comment 1.6

5. The discussion section touches on an important concept—the notion of echo chambers—which might be worthy of more in-depth exploration. It would be beneficial if this could be expanded upon, perhaps by differentiating between "echo chambers," influenced by homophily, and "filter bubbles," induced by feed algorithms.

Author Response 1.6

The reviewer touches on an important point which we are happy to build on and clarify in the revised manuscript. In the revised text we have expanded the text on echo chambers and have also distinguished between echo chambers and filter bubbles, an important distinction given that our engagement results may be influenced by the down ranking of toxic content by Twitter's feed algorithm, **see lines 710 - 726**. We have also cited additional studies to contextualize our results, see **Refs. [30, 74 - 76]**.

Reviewer Comment 1.7

In closing, I am excited about the potential impact of this research and look forward to seeing how it can be refined to tell a story that's not only informative but also compelling and coherent. I believe with these suggested improvements, the manuscript will be well-suited for publication.

Sincerely,
Almog Simchon

Author Response 1.7

We are very grateful to Dr Simchon for his enthusiastic and constructive review. Thanks again for your efforts – your comments have helped us substantially improve the paper.

Response to Reviewer 2

Reviewer Comment 2.1

This uses a large Twitter dataset to investigate the extent to which affective and interactional polarization align in several countries. The work is very relevant, as this is still an open question, and the methodology is mostly sound. There are several interesting and new results on the toxicity and engagement within and across ideologies, however, there are also several limitations to their results, such as the fact they studied only 24h of Twitter activity and only looked at activity related to elected politicians. As the authors recognize, this is a first step to understanding this issue better. However, it is not definitive evidence showing that affective and interactional polarization are always aligned. For this reason, in its present form, the paper may fall just a bit short in terms of the generalization of its results.

Author Response 2.1

We thank the reviewer for their efforts reading our manuscript and for their detailed comments and positive feedback.

As the reviewer has acknowledged, we do not claim that our results generalize across all countries and instances – in fact, we would argue that no social media research can claim to generalize across all instances. This is precisely why we argue that it is critical to study polarization across multiple countries, and to not assume how results will translate to other contexts. Our work aims to highlight common trends across countries, but acknowledges important exceptions and caveats. These points are discussed in **lines 115 - 118**, **lines 787 - 806** and **lines 819 - 822**.

Below, the reviewer will see that we have included detailed responses to each of their comments and a number of new results which, we hope they will agree, have substantially improved the manuscript.

Reviewer Comment 2.2

I have two main issues:
1) The authors decided to focus only on tweets related to elected politicians, but I expect that a lot of political conversations occur without mentioning or retweeting politicians. For example, it would have made sense to include tweets from media outlets as well. Could the authors repeat their analysis by including the most prominent media outlets in each country or at least provide a clear reason for not explicitly including them?

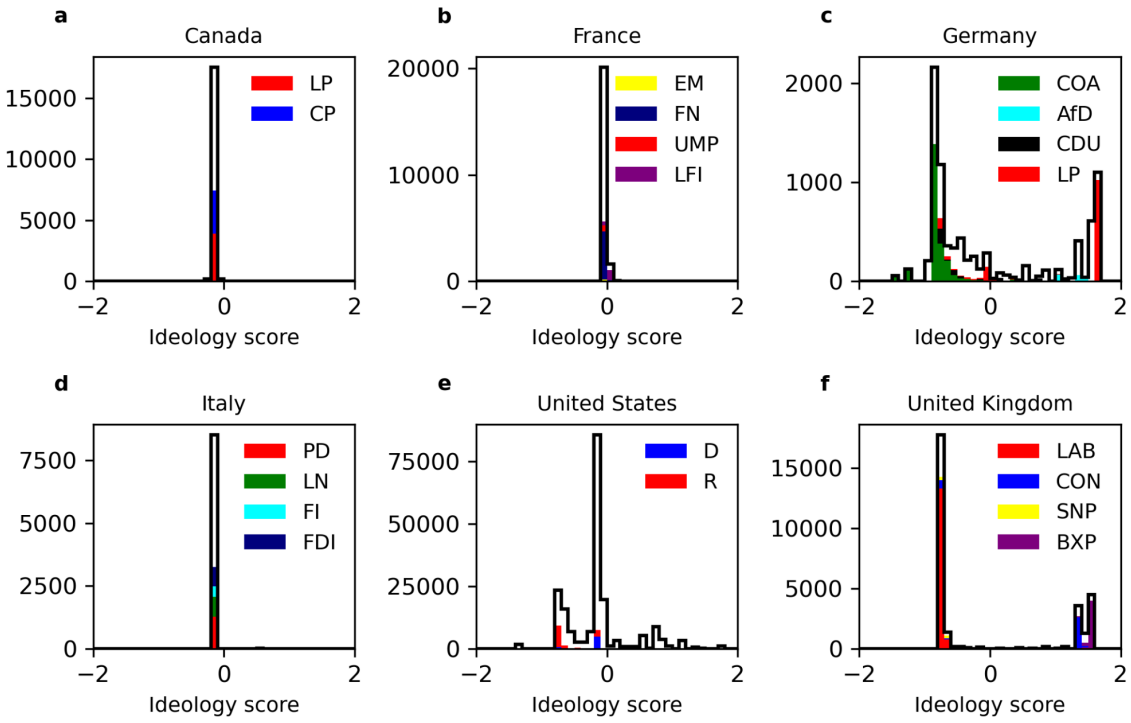
Author Response 2.2

We thank the reviewer for this important comment which we are happy to address. Firstly, we would like to clarify that our analysis of toxicity and engagement does not only include tweets which mention elected politicians. Rather, we use retweets of elected politicians to identify Twitter

users with a political ideology, and then we consider any mention between this cohort of users for the toxicity and engagement analysis – these tweets do not need to include a mention of a politician. We have clarified this point in the revised text, **see lines 160 - 165**.

Regarding the inclusion of media outlets, there are strong empirical reasons why, in our opinion, media outlets should not be included in the set of influencers which users retweet in order to have an ideological score assigned to them. Firstly, we only have data for news outlets in six of the nine countries studied. Secondly, although news outlets are often identified as having a political leaning (right, left, or centrist), in many countries they do not have a clear partisan affiliation. This is important given the distinction in the affective polarization literature between political identity and political ideology (see papers by Lilliana Mason). Third, recent literature suggests that the majority of social media users have a politically moderate news diet, which, if included in our set of influencers, may suppress the identification of interactional polarization between opposed poles and may give the appearance of a unimodal ideology distribution (see https://www.science.org/doi/10.1126/sciadv.abn9418 , https://journals.sagepub.com/doi/10.1177/2158244019832705 , https://onlinelibrary.wiley.com/doi/10.1111/ajps.12589). This is the case for three of the six countries where we have news outlet data on Twitter, see figure below. Finally, while it is highly probable that interactions involving elected politicians are political in nature, this is not the case with interactions targeting news outlets. Specifically, recent literature shows that the majority of interactions with news domains online are not political; more than half the news users consume is apolitical (e.g., relating to food recipes or sports; see https://journals.sagepub.com/doi/full/10.1177/19401612211047194). This means that including news domain interactions may conflate political interactions with apolitical interactions.

As evidence of point 3 above, we have recomputed the latent ideology (scores are not rescaled) using elected politicians and news domains (those listed in our NewsGuard dataset) for the six countries where we have news outlet data. This is shown here:



The latent ideology computed using a combined dataset of influencers including elected politicians and news domains for the six countries where we have available news domain data. *Ideological scores are not rescaled. In each subfigure, the histogram outlined in bold shows the number of Twitter users with an ideology score in the binned range. Colored bars show the modal political party of users in the binned ideology range. Area in white corresponds to users without a unique modal political party, including users who primarily interact with news media outlets.*

The above figure shows that the inclusion of news domains in Canada, France, and Italy collapses the multipolar ideology distribution derived in the main paper onto a unimodal distribution, since interactions with politically moderate news domains obscure the structure of explicitly political interactions with elected politicians. In Germany and the United Kingdom, the inclusion of news domains does not appear to significantly impact the derived ideological distribution. In the United States, part of the bipolar distribution of political interactions is retained when including news outlets in the set of influencers. However, the ideological distribution now includes regions which are dominated by news interactions, not political interactions.

For these reasons, and to ensure consistency across the nine countries we analyze, we have retained our original methodology using elected politicians as our set of influencers for calculating the ideological score of users. We stress however that our toxicity and engagement analysis does include interactions not mentioning an elected politician, **see lines 160 - 165**, and we have revised the manuscript to explain why news domains are not included in the set of influences, **see lines 165 - 175**. To support this discussion we have added multiple new references, see **Refs. [59 - 62]**.

Note that in response to a comment from Reviewer 1, we have added an analysis comparing political and apolitical interactions showing that political interactions (in- and out-group combined) are more toxic than apolitical interactions in all nine countries studied, **see lines 574 - 604**.

Reviewer Comment 2.3

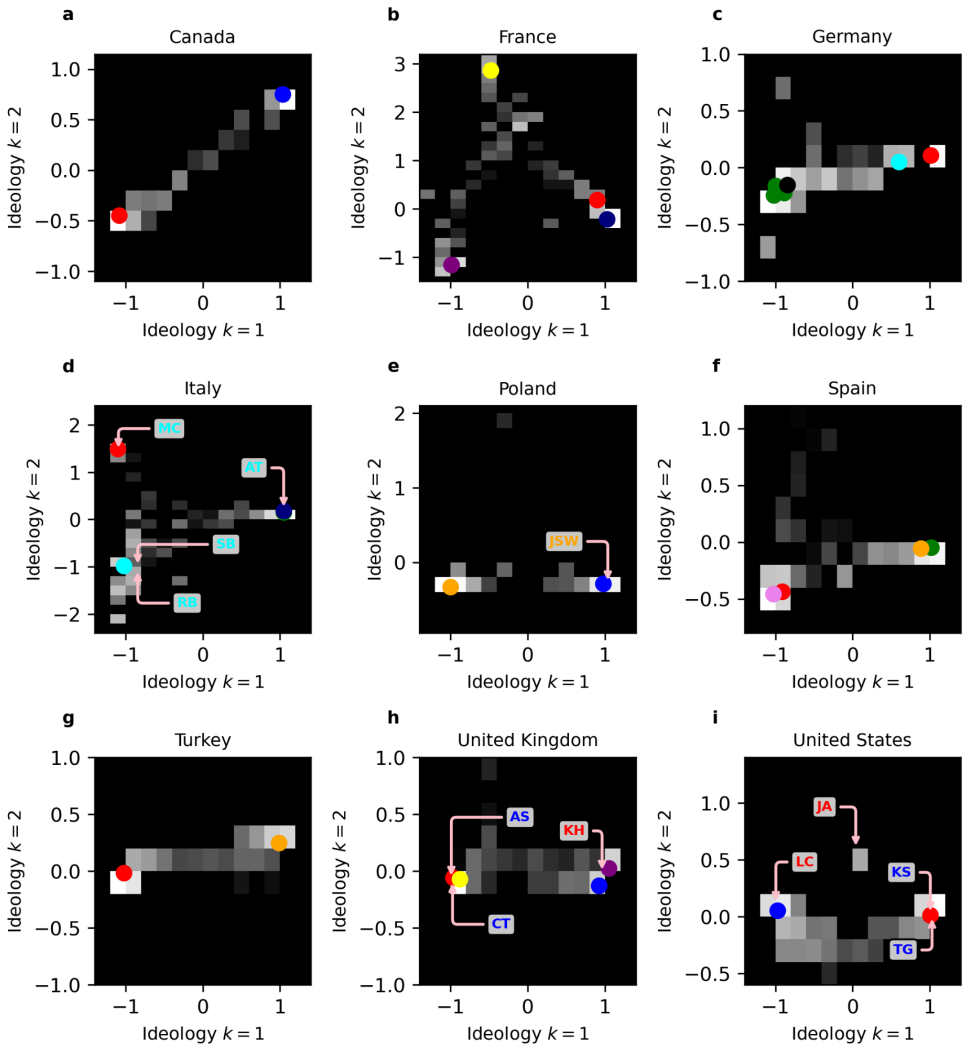
2) About the ideology scores extraction. This is where I have some questions about the methodology.
First, the authors claim that they find two opposing groups in each country, and much of the analysis relies on this. But, they actually force the ideology to be one-dimensional. The ideology is extracted from an SVD decomposition of a rescaled bipartite incidence matrix, and only the first dimension is used. Here, the authors could look at the other dimensions to see if they have significant information (the corresponding singular value should give an idea of the importance of the dimension). In particular, looking at Fig. 2. it seems that France, Germany, Poland and others may be better captured with more dimensions.

Author Response 2.3

The reviewer raises an important point which we thank them for. Firstly, it is important to stress that our claim is that the latent ideology identifies the *primary* structural divide in a country's political communication network.

The choice to focus only on the first dimension of the SVD decomposition was made since it allows for a consistent methodology to be applied across the nine countries studied. This is of course not without its limitations, since in multi-party contexts it groups parties into the political left or the political right, a limitation we state in **lines 356 - 363**. It is our ambition to improve on these methods to better account for the multi-party nature of some countries in the future, however, this is far from a trivial exercise and would require extensive work to assess how to fairly measure differences in polarization between countries with varying numbers of parties of different sizes. We have stated this ambition for future work in **lines 747 - 769**.

To address the reviewer's suggestion regarding the second dimension of the SVD decomposition, we have recomputed the latent ideology using the first two dimensions. We then plot a heatmap corresponding to the density of users in each of the first two dimensions of the SVD decomposition, and have marked the median position of specific political parties. This is shown here:



Density plots showing the latent ideology for the political interaction network of each country computed using the 1st (k=1) and second (k=2) dimension of the SVD decomposition. *High user density regions in white, low density in black. Density color shown on a logarithmic scale. Colored points show the median position of political parties. Initials in gray boxes mark the position of select politicians (see Response 2.7). Note that because our standard normalization method requires two peaks in the ideology distribution, and because there are several countries with only a single peak in the k=2 ideology, we have not applied our standard normalization to the k=2 dimension.* *Canada: LP: Liberal Party (red), CP: Conservative Party (blue). France: EM: En Marche (yellow), FN: Rassemblement National (navy), UMP: Les Républicains (red), LFI: La France Insoumise (purple). Germany: COA: Coalition (green) (Sozialdemokratische Partei Deutschlands / Freie Demokratische Partei / Bündnis 90/Die Grünen), AfD: Alternative für Deutschland (light blue), CDU: Christlich Demokratische Union Deutschlands / Christlich-Soziale Union in Bayern (black), LP: Die Linke (red). Italy: PD: Partito Democratico (red), LN: Lega Nord (green), FI: Forza Italia (light blue), FDI: Fratelli d'Italia (navy). Poland: PiS: Prawo i Sprawiedliwość (blue), PO: Platforma Obywatelska (orange). Spain: PSOE: Partido Socialista Obrero Espanol (red), VOX: Vox (green), UP: Podemos (pink), Cs: Ciudadanos (orange). Turkey: AKP: Adalet ve Kalkınma Partisi (orange), CHP: Cumhuriyet Halk Partisi (red). United States: D: Democrats (blue), R: Republicans (red). United Kingdom: LAB: Labour (red), CON: Conservative (blue), SNP: Scottish National Party (yellow), BXP: Brexit Party / UK Independence Party / Reform (purple).*

The figure shows that in the majority of countries studied (Canada, Germany, Poland, Spain, Turkey, United Kingdom, United States), using exclusively the k=1 dimension for the latent ideology does an excellent job of capturing the core structure of a country's political interaction

networks. This includes countries with a strong divide between two key parties (Canada, Poland, Turkey, United States), but also some multi-party contexts, most notably Germany, Spain, and the United Kingdom. This emphasizes that, even in a multi-party system, using only the 1st dimension of the SVD decomposition does an excellent job of capturing the main structure of political interaction networks on Twitter.

There are two countries where the second dimension of the SVD decomposition does reveal additional important details regarding online political interactions: Italy and France. In the case of Italy, it is important to note that the period studied corresponds to a pre-election period with significant political flux. Here, the primary divide in the k=1 dimension is between centrist parties and far-right parties. The k=2 dimension does not split far-right parties, but does reveal a separation between the center left and the center right.

In France, similar differences are observed. Here, the primary divide in the k=1 dimension is between centrist and leftist parties on the left of the political spectrum, and right / far-right parties on the right of the spectrum. The k=2 dimension again does not separate the rightist parties, but reveals significant divides between Macron's centrist *En Marche,* and the leftist party *La France Insoumise* led by Melanchon.

Using only the k=1 dimension in these two instances reduces the ideological spectrum to centrist parties vs. the far-right in Italy, and left/ centrist parties vs. the right / far-right in France. We acknowledge that there may be considerable debate as to whether this *primary* divide is a fair reflection of the national politics of each country. However, as a first approximation, and in the absence of advanced methods for comparing levels of affective polarization in two-party contexts rather than in multi-party contexts using structures of interactional polarization, we are of the view that our choice to use exclusively the k=1 dimension is acceptable.

To ensure maximum clarity in the revised manuscript, we have significantly expanded our discussion of the limitations of only using the k=1 dimension, but have discussed why we believe it is justified as a first approximation, **see lines 309 - 315**. We have also added the above figure to the supplement to ensure the reader is better able to identify the structural divides present in various countries, **see Figure S8**. This new result is referenced in the main manuscript in **lines 253 - 256** and **lines 907 - 917**.

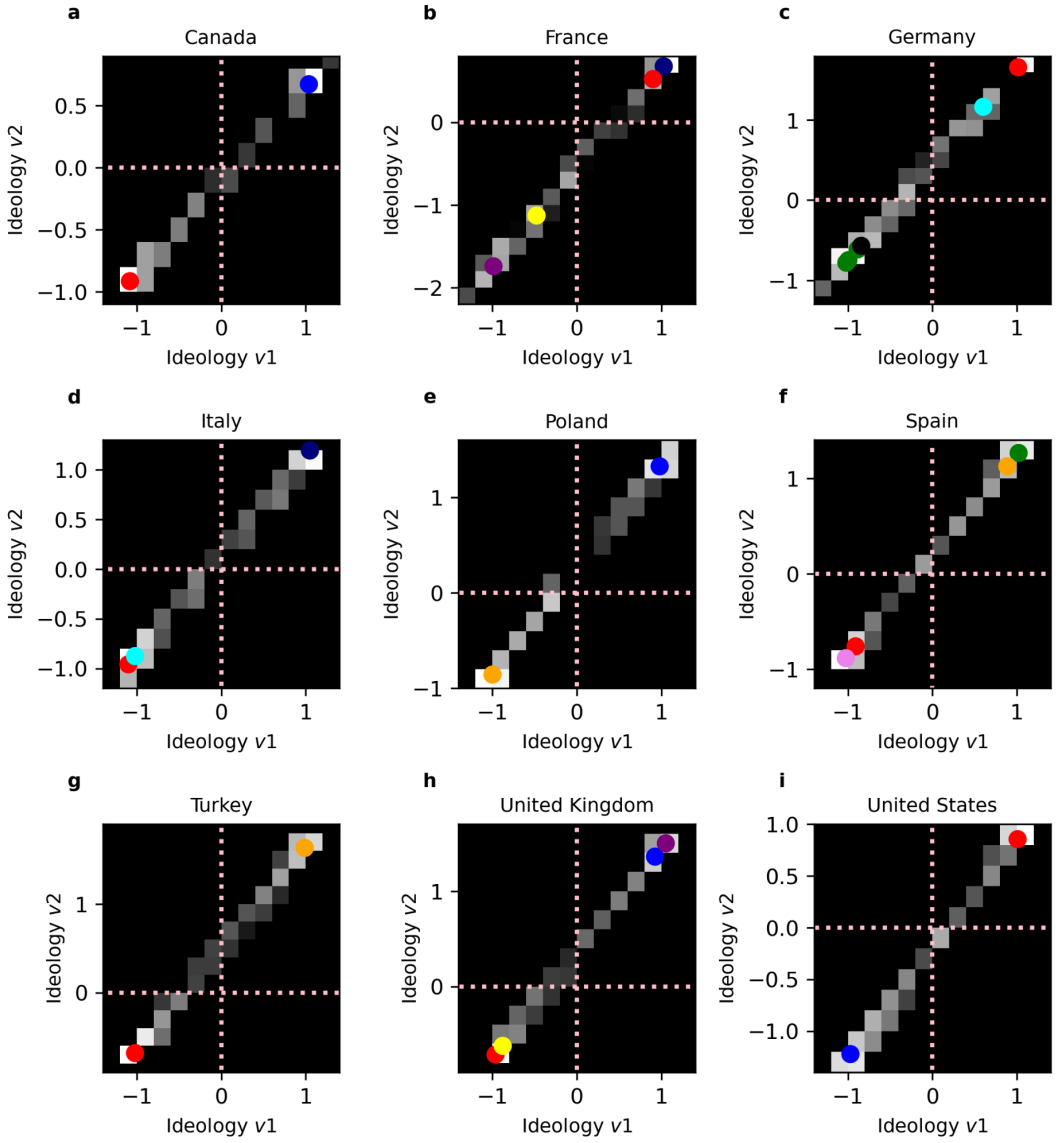
Reviewer Comment 2.4

Then there is the question of the rescaling using the two largest peak. There could be other ways to scale the scores (e.g., by removing the mean, for example), which would change where the 0 value is and therefore impact which users are considered left or right. This probably would not affect the results too much, but some robustness checks would be good.

Author Response 2.4

We thank the reviewer for this question. To assess the robustness of our results, instead of rescaling such that the two primary peaks align with -1 and +1 on the ideological scale, we have tested an alternate rescaling where we take the raw ideological scores, subtract the mean, and then normalize values by dividing scores by the distribution's standard deviation. We then plot these new ideological scores against the original ideological scores where the primary peaks

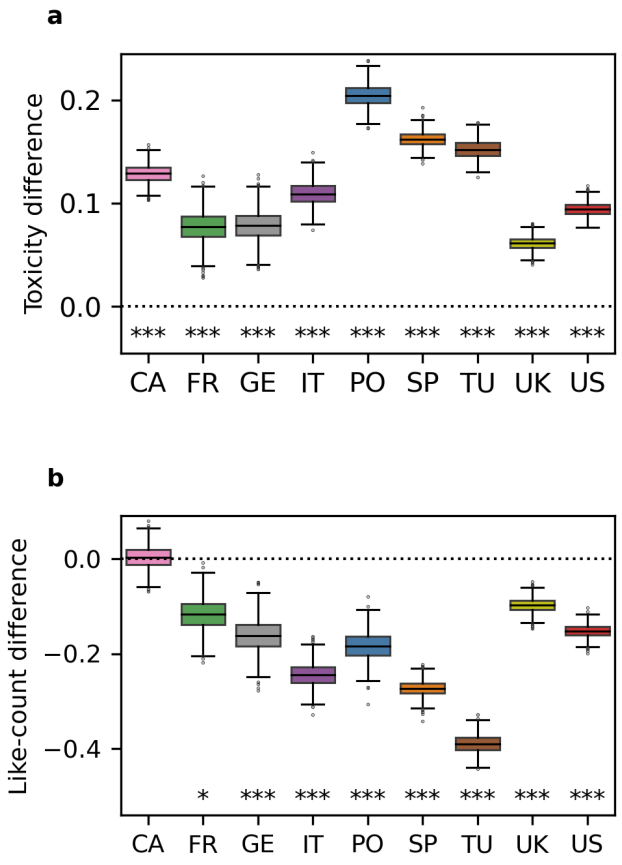
align with -1 and +1. As in the original paper, for both rescalings, we label the political left as all users with an ideological score less than zero, and the political right as all users with an ideological score greater than zero. This is shown here:



Density plots showing the latent ideology for the political interaction network of each country using the original rescaling ("Ideology v1"), and using the new rescaling where raw ideological scores have the mean subtracted, and scores are then normalized according to the distribution's standard deviation ("Ideology v2"). *The political left are users with an ideological score less than zero. The political right are users with an ideological score greater than zero; this crossover point is marked with the pink dotted lines. Users in the bottom left and top right quadrants of the density plots have the same political label (i.e., left or right) using either rescaling. Users in the off-diagonal quadrants have a different political label using the two different rescalings. The median position of politicians from each political party in all nine countries is unaffected by the choice of rescaling, and only a small number of users change label when using an alternate rescaling.* *Canada: LP: Liberal Party (red), CP: Conservative Party (blue). France: EM: En Marche*

(yellow), FN: Rassemblement National (navy), UMP: Les Républicains (red), LFI: La France Insoumise (purple). Germany: COA: Coalition (green) (Sozialdemokratische Partei Deutschlands / Freie Demokratische Partei / Bündnis 90/Die Grünen), AfD: Alternative für Deutschland (light blue), CDU: Christlich Demokratische Union Deutschlands / Christlich-Soziale Union in Bayern (black), LP: Die Linke (red). Italy: PD: Partito Democratico (red), LN: Lega Nord (green), FI: Forza Italia (light blue), FDI: Fratelli d'Italia (navy). Poland: PiS: Prawo i Sprawiedliwość (blue), PO: Platforma Obywatelska (orange). Spain: PSOE: Partido Socialista Obrero Espanol (red), VOX: Vox (green), UP: Podemos (pink), Cs: Ciudadanos (orange). Turkey: AKP: Adalet ve Kalkınma Partisi (orange), CHP: Cumhuriyet Halk Partisi (red). United States: D: Democrats (blue), R: Republicans (red). United Kingdom: LAB: Labour (red), CON: Conservative (blue), SNP: Scottish National Party (yellow), BXP: Brexit Party / UK Independence Party / Reform (purple).

The figure shows that the choice of rescaling has a minimal effect on whether a user is labeled as a member of the political left or right. To ensure full robustness, we use these new political labels derived using the second rescaling and recompute the difference in out-group vs in-group toxicity, and the difference in out-group vs in-group engagement for all nine countries (equivalent to the original Figure 3, but with the new rescaling). This is shown here:



Out-group interactions are more toxic, but receive lower engagement, than in-group interactions. Out-group and in-group interactions are determined using the alternate rescaling ("Ideology v2"), instead of the rescaling used in the original paper ("Ideology v1"). *(a) Boxplots for the bootstrapped difference between the median out-group interaction toxicity, less the median in-group interaction toxicity for each of the nine countries. (b) Boxplots for the bootstrapped difference between the mean log likes received by an out-group post relative to the mean log likes received by an in-group post. Boxplots show the bootstrapped median, interquartile range (IQR), whiskers for 1.5 times the IQR from the hinge, and points for outliers*

*(see Methods). Stars indicate statistical significance using the non-parametric Mann-Whitney U test on full distribution (see Methods): p<0.05: *, p< 0.01: **, p<0.001: ***.*

The figure looks almost identical to the original Figure 3 (although we do note there are subtle differences). However, the sign of the toxicity and engagement differences and the level of statistical significance is unchanged for all nine countries and both measurements. These robustness figures have been added to the supplement, see **Figure S6** and **Figure S7** and are noted in the revised text in **lines 375 - 377** and **lines 925 - 926**.

Reviewer Comment 2.5

Smaller remarks:
- line 333: " Figure 3(a) shows boxplots of the bootstrapped difference in the median toxicity quantile of out-group interactions less the median toxicity quantile of in-group interactions for each country". This sentence is confusing. The median is a quantile, but I don't see what the median toxicity quantile is... Do the authors mean they looked at the ranks of posts sorted by toxicity? Maybe a mathematical definition would clarify things.

Author Response 2.5

We thank the reviewer for highlighting this ambiguity which we are happy to clarify. The confusion here arises because the figure shows the difference between the median quantile for in-group interactions and the median quantile for out group interactions, not the quantiles themselves, with the toxicity quantile assigned a number between 0 and 1 (as opposed to percentiles which run from 0 to 100). The individual in-group and out-group quantiles (not the difference) for the nine countries are shown in the supplement in **Figure S3**.

Taking the example of Poland, **Figure S3** shows that the median toxicity quantile for in-group interactions is 0.4 and the median toxicity quantile for out-group interactions is approximately 0.6. Therefore, taking the difference of the median quantiles gives 0.6 - 0.4 = 0.2. This is the value that is shown for Poland in **Figure 3a**.

In the revised manuscript, we have altered the text to clarify that the toxicity quantiles are defined between 0 and 1, have given an example of how the difference in the median quantile is computed, and have more explicitly noted that the raw toxicity quantiles for in- and out-group interactions can be seen in **Figure S3** of the supplement. See **lines 390 - 395** in the revised manuscript. Note also that following a request from reviewer 4, see **Response 4.3**, we have added an additional figure to the supplement, **Figure S4**, which shows the distribution of raw toxicity scores for the nine countries which further confirms that out-group toxicity is consistently higher than in-group toxicity.

Reviewer Comment 2.6

for the engagement study, why did they use the mean log like-count and not the median, as for the toxicity?

Author Response 2.6

We thank the reviewer for raising this question. In the case of the engagement study, it is important to note that posts receive an integer number of likes (0, 1, 2, … etc.) and that the distribution of the like-count is fat-tailed, i.e., there are vastly more posts with a small number of likes than there are posts with a large number of likes. In our case, between 66% (Poland) and 78% (USA) of the posts analyzed across the nine countries in Figure 3 receive zero likes. This means that computing the median like-count, or using like-count quantiles, is ineffective because the median like-count across countries is always zero. For this reason we prefer to use the mean rather than the median for the like-count distribution. However, statistical tests often perform poorly when applied to fat-tailed distributions. For this reason, it is generally a good rule of thumb to take the logarithm of a fat-tailed observable (such that it is no longer fat-tailed) prior to computing statistics on it, hence why we use the logarithm of the like-count. Note, that these issues do not affect the toxicity analysis since toxicity scores take continuous values between 0 and 1. In the revised manuscript we have clarified this choice in **lines 460 -467**.

Reviewer Comment 2.7

In the conclusion, "our results also reveal how partisan non-conformists are essentially treated as members of the political opposition." I wonder if this is not because they only used a 1-D ideology. Including more dimensions could capture the ideology of these non-conformists.

Author Response 2.7

We thank the reviewer for this interesting question. To address this comment, we have marked the position of key partisan non-conformists mentioned in the original paper from select countries on the figure comparing the k=1 and k=2 dimensions of the SVD decomposition shown in **Response 2.3** above. We have also highlighted a few additional politicians which were not previously discussed and which highlight interesting details in the political interaction network.

The figure confirms that in the USA, Liz Cheney (LC) aligns with the Democrats, Tulsi Gabbard (TG) aligns with the Republicans, and Kyrsten Sinema (KS; not mentioned in the original paper) aligns with the Republicans. Interestingly, the only politician who appears to sit outside the primary Democrat Republican divide is Justin Amash (JA) who was previously a Republican, but now sits in the House of Representatives as the only Libertarian. In the UK, the figure confirms that Conservatives Anna Soubry (AS) and Charles Tannock (CT) both align with the left, and Labour politician Kate Hoey (KH; not mentioned in the original paper) aligns with the right. In the latter case this result is expected given that Kate Hoey was one of the most prominent Labour party politicians to support Brexit. Finally, in Poland Jacek Saryusz-Wolski (JSW) aligns with Prawo i Sprawiedliwość, reflecting his previous defection from Platforma Obywatelska.

In Italy, the nature of the outliers highlighted is slightly different, as revealed by the second dimension of the SVD decomposition. As noted in the original text, the period studied corresponds to just before the Italian elections where Forza Italia allied itself with Giorgia Meloni's Fratelli d'Italia. However, this alliance was only agreed after the election. In the figure, we see that FI politicians appear at various points in the political spectrum. In particular, Antonio Tajani (AT), who would later become Meloni's vice President, already aligns with the FDI at this point.

Conversely, FI politicians who would only later join the FDI in Government (Renato Brunneta, RB, and Silvio Berlusconi, SB) are found in one cluster on the left of the ideological spectrum, and FI politicians who would defect from the party and join centrist parties are found in a different cluster on the political left (Mara Carfagna; MC).

In the revised manuscript, we have updated the discussion of political non-conformists to reflect these new results, see **lines 334 - 338**.

We thank the reviewer for their detailed comments which have helped to substantially improve the manuscript.

Response to Reviewer 3

Reviewer Comment 3.1

The authors analyze a quite unique dataset comprising all public tweets and re-tweets in one day across countries. The authors start by applying an “unsupervised learning” technique to derive an ideological score for each user and cluster them based on interaction patterns with active politicians on Twitter. This ideological score is used to distribute users between two groups — political left and political right. Then, the authors move on to study the level of segregation revealed according to this group definition; the level of toxicity and engagement in in/out-group posts; the level of reliability in sources referenced by the two groups; and other characteristics of Twitter interactions, independent on group identity (e.g., political vs apolitical interactions and reciprocity in interactions).

The authors conclude that Twitter reveals high levels of interactional polarization: users belonging to different ideological group preferentially retweet politicians from different political parties. The authors conclude that out-group interactions are more toxic and less engaging (that is, receive less likes). Finally, the ideological group associated with the political right references less reliable media sources.

Overall, I found the analysis presented in this manuscript sound. I believe the conclusions are original and interesting to a multidisciplinary community interested in dynamics of political discourse on social media platforms. Although the paper reports many findings, the results section follows a very clear structure, and it is easy to follow all steps of the analysis. My general assessment is positive. Nonetheless, I have some suggestions for improvement.

Author Response 3.1

We are grateful to the reviewer for their time and effort and thank them for their positive and constructive comments.

Reviewer Comment 3.2

The definition of “political” and “apolitical” content followed here ignores posts that might be politically loaded without mentioning politicians’ accounts. The authors compute users’ ideological score given their retweets of politician’s accounts; however, users can express their views or endorse/criticize policies without explicitly referring to a politician. Alternative methods such as topic modelling could provide ways of classifying the ideological leaning of posts without resorting to mentions to the accounts of elected politicians. The authors should discuss eventual limitations of their definition of political content.

Author Response 3.2

The reviewer raises a very important point which we are happy to address. Having looked at our data more closely, we agree that the original manuscript was insufficiently clear as to how we used the terms political and apolitical.

In the revised manuscript we have made a major effort to improve this clarity by stating that the term “political” is used to refer to mentions between accounts who are explicitly engaged with elected politicians, whereas the term “apolitical” is used to refer to all other mentions. As part of this, we acknowledge that the “apolitical” cohort will contain some content and interactions which are political in nature, but, unlike the “political” cohort, will contain a large amount of content and interactions which are clearly not political in nature. This language is now explicitly explained in the revised manuscript, **see lines 538 - 545**, and is further emphasized in the limitations section, **see lines 696 - 707**.

Note that following suggestions from Reviewer 1, we have added a new analysis to the paper where we compare political interactions (i.e., interactions between two accounts who both have an assigned political ideology) to apolitical interactions. This analysis shows that apolitical interactions are less toxic than political interactions in all nine countries, **see lines 574 - 604** and the new **Figure 6**.

Finally, regarding the reviewer’s suggestion that topic modeling may be used to assign a political leaning to posts. In previous papers we have made extensive use of topic modeling to understand the content of political Twitter interactions (see for example https://www.nature.com/articles/s41558-022-01527-x or https://academic.oup.com/pnasnexus/article/2/11/pgad346/7329980). However, in both instances that analysis only involved English language content, and we could only classify (with moderate reliability) posts which were relatively long. Additionally, when analyzing these topics we did extensive manual work to refine topics and minimize duplicates, and even then less than half of content was meaningfully classified as belonging to a specific topic. Finally, in many cases we observed that topics were common to both the political left and right, and that the leaning of a post could not be understood in the absence of the context of the posts being referenced. For these reasons, it is our opinion that assigning ideological scores to content using topic modeling is not really feasible in the current paper, especially because our team does not speak all the necessary languages analyzed, and because our dataset is too large to allow for a robust checking of topic accuracy. Rather, we prefer to explicitly acknowledge the limitations of the definition of political and apolitical, **see lines 778 - 784**, and have stated in the discussion that topic modeling may be an interesting strategy to pursue in the future for understanding political twitter interactions, **see lines 784 - 786**. We hope the reviewer will find this acceptable in the current instance, but thank them for the suggestion.

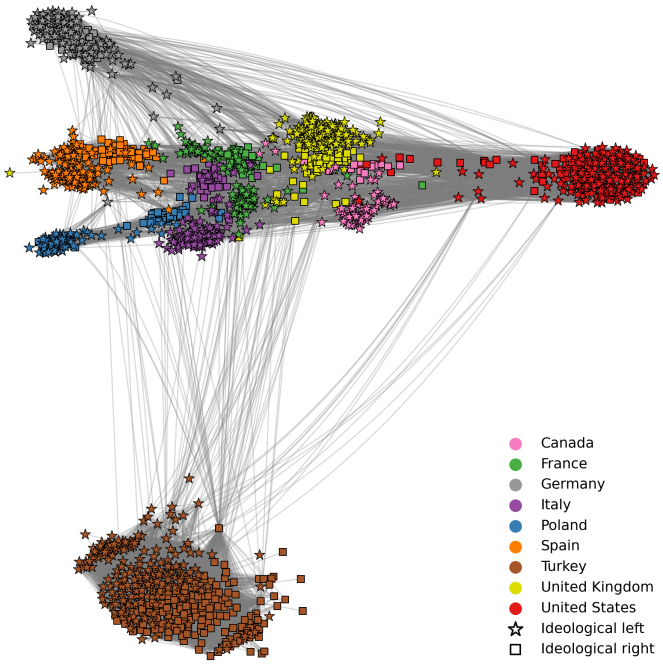
Reviewer Comment 3.3

The decision to ignore users that retweet more than 10 politicians (why 10?) and the decision to add use 2 retweets (why 2?) as the minimum threshold to add a link in the co-retweet network still seems rather arbitrary. The authors could provide extra data on the robustness of their conclusions if those two thresholds change.

Author Response 3.3

We thank the reviewer for highlighting this issue. It is important to stress that these arbitrary thresholds are only used for the network visualization in figure 1 for the purposes of visual clarity. For all other analyses throughout the paper (figures 2 to 6), no minimum or maximum retweet thresholds are imposed.

In the case of figure 1, the thresholds are included to ensure the interpretability of the network visualization. Since the network visualization is included for qualitative analysis only, we deemed this choice to be acceptable, since without these thresholds there are (1) too many edges between politicians for a clear network visualization, and (2) retweets by highly active accounts who retweet a very large number of politicians result in some spurious connections between politicians who would otherwise not be connected in the visualization. Note that of the 33,000 Twitter users who retweet two or more politicians across the 9 countries studied, 96% of those users retweet 10 or fewer politicians. However, if the retweets of the 4% of users who retweet more than 10 users are not excluded, edges between politicians as a result of these highly active users are responsible for 86% of all connections in the co-retweet network. For this reason, and for visualization purposes only, we chose to exclude these highly active retweeters. However, had we retained these highly active accounts, the network visualization would look like this:



Qualitatively, this figure has a similar structure to the co-retweet network shown in **Figure 1** of the main paper. However, there are many more edges between nodes, and nodes are bunched together in much smaller clusters so that groups are harder to identify. It is for this reason we opted for the sparser network visualization shown in **Figure 1**.

In the revised text, we explicitly clarify that the minimum and maximum retweet thresholds apply to the network visualization only, and clarify why this choice has been made, see **lines 215 - 219** and **lines 852 - 855**.

Reviewer Comment 3.4

The dataset used is distinctive as it comprises all Twitter posts in a 24-hour period. At the same time, this 1-day window might pose limitations. September 21 2022 might represent a different day in terms of the electoral cycles and specific political events in different countries. As the authors discuss, the results for Germany are very much dependent on the government's position on the Russia-Ukraine war (would the same conclusions hold if the the day was September 21 2021 or September 21 2023?). It remains unclear how the results presented generalize to longer time-windows; the authors could perhaps discuss this.

Author Response 3.4

The reviewer raises an important point here – the 24 hour period of our dataset is certainly a key limitation of our study as we stated in the original version of the manuscript. In the original manuscript we discussed this limitation briefly, but acknowledge we could have gone further. In the revised manuscript we have expanded our discussion of this limitation to be more explicit about the possible pitfalls of this approach, see **lines 787 - 799**. However, we also note that our ability to identify consistent trends across countries from only 24 hours of data also highlights the strength of our approach and dataset. In the revised manuscript, we have discussed how possible future work may investigate some of the limitations of our current dataset by, for instance, tracking changes in the political interaction networks over time. Similar analysis has previously been carried out by Baqir et al. 2023 (**Ref. [80]** in the paper) which shows how Twitter interactions networks follow the structure of coalition governments in Pakistan. This work is discussed in **lines 799 - 809**.

Reviewer Comment 3.5

Minor points:
Line 657 and Figure 1 caption: It is mentioned that the network visualization is produced using a force directed drawing algorithm (ForceAtlas2). This might sound rather cryptic for readers not familiar with this algorithm. My suggestion is to provide an intuition for what this means in terms of attraction and repulsion between nodes.

Author Response 3.5

We thank the reviewer for this useful suggestion which we are happy to include in the revised manuscript. In the new methods, we give an intuitive explanation of how the force atlas algorithm works and the physics-based principles which determine the network layout. Specifically, we explain that the algorithm uses repulsive forces between pairs of nodes to push nodes apart, but includes attractive forces between nodes if they are connected by an edge, see **lines 859 - 862**. The figure 1 caption has been revised to refer to this new text in the methods.

Reviewer Comment 3.6

Line 693: “retweeted users” or “retweeted influencers”?

Author Response 3.6

We are very grateful to the reviewer for spotting this error. The text should refer to the *n* users who retweet the *m* politicians. We have revised the text to correct the error and improve clarity, see **lines 902 - 904**.

Reviewer Comment 3.7

Line 687: please clarify the dimensions of vectors r and c and the exact product implied by rc — this was confusing as r and c do not have the same dimensions.

Author Response 3.7

Thank you for raising this question. The product **rc** denotes the outer product of the column vector **r**, which corresponds to the sum over the rows of the matrix **P** and therefore has dimensions *(1, m+1)*, with the row vector **c**, which corresponds to the sum over the columns of the matrix **P** and therefore has dimensions (*n, 1)*. Here, *n* is the number of retweeters, and *m* is the number of politicians (*m+1* when including the dummy politician). The outer product **rc** is therefore a matrix of dimensions (*n, m+1).*

We have updated the mathematical description of the latent ideology in the methods to clarify these details, see **lines 887 - 896**.

Reviewer Comment 3.8

The ideological scores considered are continuous, yet they are collapsed onto a binary classification (left-right). The authors could comment potential limitations for this approach, especially in countries where many users have a score close to 0 (see France).

Author Response 3.8

We thank the reviewer for this comment. In the traditional literature on ideological polarization we agree that a binary classification of ideological scores on a left-right spectrum would be overly simplistic. However, in the context of social media scores where the ideological scores are based on interaction patterns and not policy positions, it is not clear whether it is possible to derive more nuanced ideological spectrums. For this reason, we opt to use the binary classification which ensures robust results and which connects well with the growing literature on partisan identity being a more important factor than political ideology in determining political attitudes towards an individual’s in- or out-group. This binary classification is a common approach when studying political interaction networks and polarization on Twitter, see **Refs. [21, 34, 43, 48]**.

Concerning how the classification affects countries with users near the 0 point on the ideological scale, this relates to a question asked by Reviewer 2 concerning the robustness of our ideological scores and toxicity results, **see Response 2.4**. There we use an alternate rescaling of ideological scores where a small number of users, particularly in France, move from one side of the ideological scale to the other, but where the primary divide remains between centrists and the political right. Recomputing the toxicity results using this alternate rescaling shows that our results are robust to this choice. Note, however, that particularly in the case of France we are conscious that a one dimensional ideological spectrum arguably oversimplifies the true structure of political Twitter interactions. We have chosen to retain this one dimensional spectrum in order to have a consistent methodology across all nine countries. However, we have added additional results to the supplement looking at the structure of political interaction networks in more detail, **see Figure S6**, **Figure S7**, and **Figure S8**, and have discussed these limitations in detail in the main text, **see lines 747 - 765**.

We are very grateful to the reviewer for their efforts reviewing our paper.

Response to Reviewer 4

Reviewer Comment 4.1

I have read the paper with great interest. Across nine studies, the authors investigate whether political networks on twitter are polarized. They found evidence for bipartite networks in all

countries, and for outgroup interactions being characterized by higher toxicity and less engagement.

Author Response 4.1

We thank the reviewer for their positive comments and for their comprehensive and constructive feedback.

Reviewer Comment 4.2

I only have a few minor comments, mostly related to expanding the theoretical contribution of the findings:

1) In the political ideology literature, there is a debate on whether only rightist show discrimination (see e.g. Balliet, D., Tybur, J. M., Wu, J., Antonellis, C., & Van Lange, P. A. (2018). Political ideology, trust, and cooperation: In-group favoritism among Republicans and Democrats during a US national election. Journal of Conflict Resolution, 62(4), 797-818). The authors briefly mention these results but I suggest to provide more details on this potential asymmetry. Overall, do rightist vs leftist show for outgroup toxicity, or it is the other way around? Or they simply show the same levels of toxicity? Is there variation across countries?

Author Response 4.2

We thank the reviewer for their interest in the results on asymmetries between toxicity from the political right and left, and for pointing us towards the very interesting paper by Balliet et al. (2018). We agree that these results are important and that they should be discussed in greater detail in the revised manuscript.

As previously noted in the supplementary material, and shown in **Figure S3d**, there are no consistent trends across countries showing that the political left or the political right are more toxic. In seven of the nine countries studied, there is no statistically significant difference in right to left toxicity relative to left to right toxicity. In the United Kingdom, the political left are statistically significantly more toxic than the right. In Poland, the right are more toxic than the left. In the revised manuscript we have discussed these results in more detail and have contextualized the results using the Balliet et al. (2018) paper, see **lines 433 - 449** and the new **Ref. [70]**. As noted in the manuscript, out-group interactions are significantly less common than in-group interactions, and when we split this already small group of interactions into right to left mentions, and left to right mentions, we are left with acceptable, albeit small sample sizes, relative to the samples used in the rest of the paper. For this reason, we have added a comment to the discussion arguing that ideological asymmetries in out-group toxicity should be a priority for future research to further clarify our results, see **lines 765 - 769**.

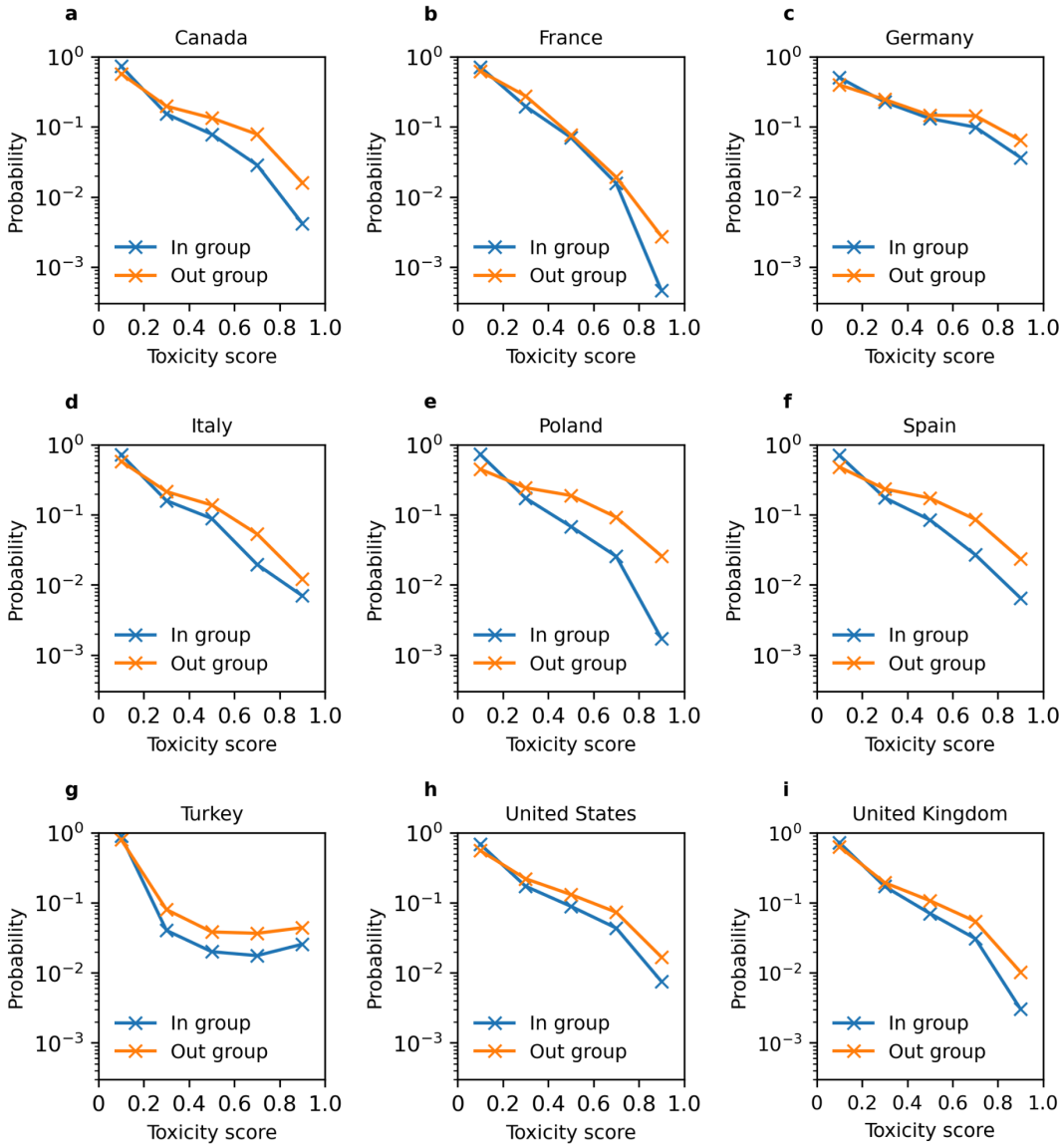
Reviewer Comment 4.3

2) The authors say that outgroup interactions are characterized by more toxicity and less engagement. Is the result on toxicity with ingroup (vs outgroup) in terms of proportion (i.e., i have

less interactions with outgroup and this lower number of interactions is mostly toxic) or absolute values? In other words, one possibility is that ingroup interactions can also be toxic but at the same time they are also characterized by many non-toxic interactions. Can the authors speak to this?

Author Response 4.3

We thank the reviewer for raising this interesting question. To address this, in the revised manuscript we show the distribution of raw toxicity scores (not just the median quantile) for in-group and out-group interactions across the nine countries. This is shown in the new **Figure S4** in the supplementary material and is referenced in the main text, see **lines 403 - 407**. For the reviewer's convenience, we include a copy of the new figure here:



The probability that in-group and out-group mentions have a toxicity score in the binned range, grouped by country. *Raw toxicity scores are binned in the ranges [0.0,0.2), [0.2,0.4), [0.4,0.6), [0.6,0.8), [0.8,1.0]. Points show the probability that a post from a given country which includes an in-group (blue) or out-group mention (orange) fall in the specified toxicity range. Across all nine countries, in-group mentions have a higher probability of falling in the low toxicity bin (scores between 0 to 0.2) than out-group mentions. Across all countries and all other toxicity bins (scores > 0.2), out-group mentions have a higher probability than in-group mentions. For the most toxic posts (scores in the range [0.8,1.0]), the probability of an out-group post falling in the binned range is between 1.7x (Germany) and 15x (Poland) higher than for in-group mentions.*

The above figure shows that, independent of how common in- and out-group mentions are, out-group mentions are consistently more toxic than in-group mentions. However, across all nine countries, the majority of both in- and out-group mentions have low toxicity scores (binned scores <0.2).

Reviewer Comment 4.4

3) The authors mention the UK as an outlier (as they show higher fraction of toxicity towards ingroup). Given the amount of data, I would actually suggest to give importance to this result. In a way, it show the strong limitation of only focusing on one country (the US). These findings show that outgroup toxicity is not a universal but at times, people can also be nasty with ingroup members.

Author Response 4.4

We thank the reviewer for highlighting this result. To clarify, in the UK out-group interactions are more toxic than in-group interactions when aggregating the tweets from the political left and the tweets from the political right. The outlier is that if we focus only on the political right, tweets authored by the right mentioning the right are more toxic than tweets authored by the right mentioning the left (i.e., in-group mentions are more toxic than out-group mentions on the right). On the left, we observe that, as expected, tweets authored by the left mentioning the right are more toxic than tweets authored by the left mentioning the left (i.e., out-group mentions are more toxic than in-group mentions on the left), and the magnitude of this difference is substantially larger than on the right such that, on aggregate, out-group interactions are more toxic than in-group interactions. The UK is the only country where such an outlier is observed.

In the revised manuscript we have clarified this point, and have followed the reviewer's suggestion noting that this is precisely why it is important to carry out comparative studies and not assume results generalize across countries, see **lines 416 - 432**. We have also added a reference to recent research, which aligns with our observations of in-group animosity, which showed that in-group animosity is sometimes as high, if not higher, than out-group animosity, see **Ref [69].**

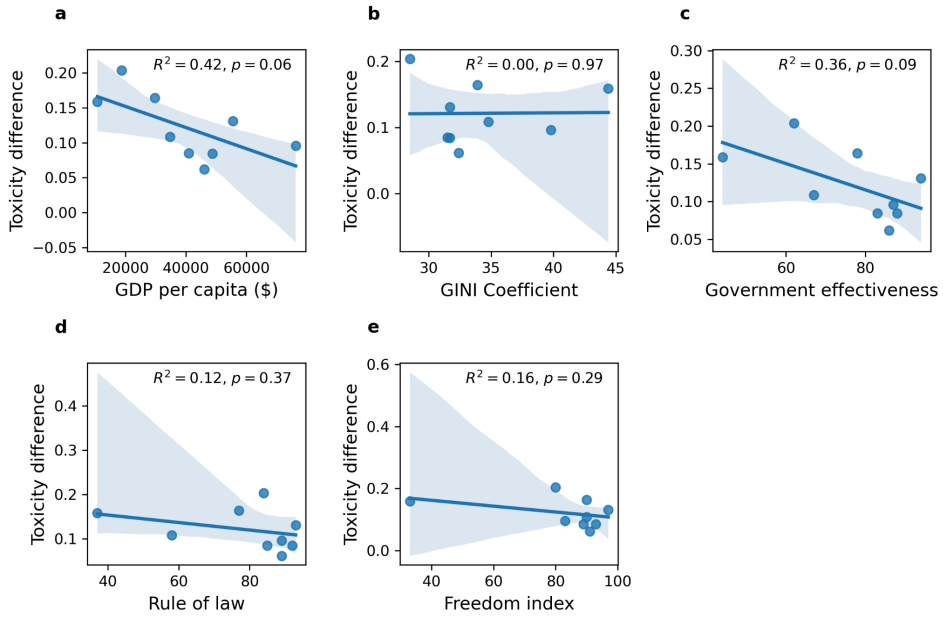
Reviewer Comment 4.5

4) Is there variation in outgroup toxicity and is this variation meaningful? Previous cross-cultural studies have tried to understand whether ingroup bias varies across societies (see Dorrough &

Gloeckner, 2016, PNAS; or Romano et al., 2021, Nat Comms). I think it would enrich the paper to have some insights on whether outgroup hate meaningfully covary with prominent cross-societal factor (e.g., gdp, or rule of law).

Author Response 4.5

We thank the reviewer for this interesting suggestion which we agree could offer additional valuable insights to our paper. To address this point, we have plotted the difference between out-group and in-group toxicity (equivalent to the values shown in Figure 3a) against a selection of prominent societal metrics for each of the nine countries studied. The resulting plots are shown here for the reviewer's benefit:



The median difference in the out-group toxicity quantile, less the in-group toxicity quantile, plotted against various important societal metrics. *Each point corresponds to one of the nine countries studied. (a) GDP per capita in US dollars. (b) The Gini coefficient. (c) Government effectiveness metric, expressed as a percentile across all countries where the metric is available. (d) Rule of law, expressed as a percentile across all countries where the metric is available. (e) The Freedom index. Data is acquired from the same sources used in [Romano et al. 2021, Nature Communications]. Figures show linear regressions, with shaded areas corresponding to the appropriate error ranges. Pearson's correlation R^2 values, and the probability, p, that the null hypothesis can be rejected (the null hypothesis states that the variables are not correlated, i.e., the slope of the linear regression is zero), are written in the top right corner of each subfigure. We do not observe a statistically significant correlation between out-group toxicity and any of the five metrics studied.*

The above figure shows that there is no statistically significant correlation between the measured difference in out-group versus in-group toxicity, and five prominent societal metrics. However, we cannot confidently state whether this result is because (1) there is no true underlying correlation

between these metrics, or (2) there is a meaningful correlation, but that with nine countries our statistical test is insufficiently powered to accurately measure any correlation.

As a rule of thumb (the “one-in-ten” rule), regressions only produce meaningful results if there are at least 10 observations per covariate. With nine countries, our dataset falls below this threshold such that we cannot be confident in our regression analysis. For this reason, and to avoid readers drawing conclusions from our results which may not be robust, we have not included this figure and its discussion in the revised paper. However, we have added to the discussion that this would be valuable additional analysis in the future if more data becomes available, see **lines 740 - 746**, and have referenced the two papers highlighted by the reviewer, see **Refs. [78 - 79]**.

We thank the reviewer for their time and effort reviewing our paper, and for their positive and constructive comments.

REVIEWERS' COMMENTS

Reviewer #1 (Remarks to the Author):

The authors have addressed all my concerns. I recommend acceptance.

Reviewer #2 (Remarks to the Author):

I thank the authors for having carefully addressed all of my issues. I am happy to recommend the publication of the manuscript in Nature Communications

Reviewer #3 (Remarks to the Author):

Let me thank the authors for the clarifications provided and comprehensive revisions introduced. The answers in their rebuttal and new content provided satisfy my previous points: 1) the distinction between political and apolitical content is now clearer, 2) the justification not to use topic modeling is reasonable; 3) I now realize that the mention activity/retweet thresholds imposed on networks are only used for visualization and, finally, 4) I am satisfied with the acknowledgement of potential limitations of considering a 24h time window. I also confirmed that all previously mentioned minor typos were properly addressed in the revised manuscript. I do not have further comments and would be happy to see this work accepted at Nature Communications.

Reviewer #4 (Remarks to the Author):

I commend the authors for having successfully attended all my comments. I think the paper is now ready for publication.